Byeongjee Kang

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Research Interests

I develop programming languages and compiler techniques for safe, efficient execution across heterogeneous systems. My experience spans from energy-intermittent microcontrollers to SIMD/multicore servers. I leverage MLIR for compiler pipelines, WebAssembly for heterogeneous deployment, and Rust for developer-facing interfaces.

Education

PhD **Carnegie Mellon University**, Software Engineering — Pittsburgh, USA
Advised by Prof. Limin Jia — Aug 2023 – present

BS **Pohang University of Science and Technology (POSTECH)**, Computer Science and Engineering & Mathematics (double major) — Pohang, Korea, Feb 2018 – Feb 2023

- GPA: 4.26/4.3 (Ranked 1st among all graduating students in class of 2023)
- Entered with presidential scholarship offered to one outstanding high school student in the year, and graduated with the highest GPA
- Early graduation by one-year (leave of absence for military duty during Jan 2020 – Dec 2021)

Professional Skills

Programming Languages: Rust, C++, WebAssembly, SML, Python, JavaScript, Scala, Maude, Futhark

Frameworks: LLVM, MLIR, React, Django, Git

Work Experiences

Microsoft Research, Research intern at Data Systems Group mentored by Kaushik Rajan — Redmond, WA, USA, May 2025 – Aug 2025

- Compiler for database intrinsics: Auto-vectorizing compiler on MLIR with support for control flow dependencies
- Developed compiler optimization passes for database intrinsics using MLIR and Polygeist, improving the performance of database intrinsic functions with control dependencies through vectorization

Kodebox, Full-stack software engineer as military service alternative — Seoul, Korea, Jan 2020 – Dec 2021

- CodeChain Foundry: Open-source blockchain engine based on composable module system
- Worked as core developer of consensus engine, implementing BLS signature aggregation scheme and designing interface for modules to communicate with consensus engine
- Zuzu: Web-based captable management software

Publications

Fray: An Efficient General-Purpose Concurrency Testing Platform for the JVM — Jan 2025
Ao Li, Byeongjee Kang, Vasudev Vikram, Isabella Laybourn, Samvid Dharanikota, Shrey Tiwari, Rohan Padhye
10.1145/3764119 (Object-oriented Programming, Systems, Languages, and Applications (OOPSLA) 2025)

GraFeyn: Efficient Parallel Sparse Simulation of Quantum Circuits — Jan 2024
Sam Westrick, Pengyu Liu, Byeongjee Kang, Colin McDonald, Mike Rainey, Mingkuan Xu, Jatin Arora, Yongshan Ding, Umut Acar
ieeexplore.ieee.org/document/10821338 (IEEE International Conference on Quantum Computing and Engineering (QCE) 2024 (Best Paper))

Narrowing and Heuristic Search for Symbolic Reachability Analysis of Concurrent Object-Oriented Systems Jan 2024
Byeongjee Kang, Kyungmin Bae
10.1016/j.scico.2024.103097 (Science of Computer Programming 2024)

Symbolic Reachability Analysis of Distributed Systems using Narrowing and Heuristic Search Jan 2022
Byeongjee Kang, Kyungmin Bae
dl.acm.org/doi/10.1145/3563822.3568017 (International Workshop on Formal Techniques for Safety-Critical Systems (FTSCS) 2022)

Preprints

WAMI: Compilation to WebAssembly through MLIR without Losing Abstraction
Byeongjee Kang, Harsh Desai, Limin Jia, Brandon Lucia
arxiv.org/abs/2506.16048

Mr3: An Execution Framework for Hive and Spark
Sungwoo Park, Seunggon Namgung, Byeongjee Kang
dx.doi.org/10.2139/ssrn.4587845

Talks

- An MLIR Dialect for WebAssembly, WebAssembly Workshop 2025, [recording]

Ongoing Research Projects

Automatic Program Rescaling and Migration May 2025 – present
Programming language and runtime support for automatically rescaling programs (including ML applications) for resource-constrained systems
- Aiming to automatically rescale programs with correctness guarantees using general approaches such as program slicing, as well as ML-specific techniques such as quantization and sparsification

Probabilistic Modeling and Analysis Tools for Intermittent System Aug 2025 – present
Probabilistic modeling of resource usage in intermittent systems

WAMI: MLIR-based Compilation Pipeline for WebAssembly Oct 2024 – present
MLIR-based compilation pipeline for WebAssembly

Honors and Scholarships

ILJU Fellowship (ILJU Academy and Culture Foundation) Aug 2023 – July 2028
- Received total of USD 120,000 over eight semesters during graduate studies

Program for Highly Dedicated Students (POSTECH) Feb 2018 – Feb 2023
- Presidential scholarship and supporting program in POSTECH offered to one outstanding freshman each year

Presidential Science Scholarship (Ministry of Education of Korea) Feb 2018 – Feb 2023
- National science scholarship offered to top freshmen students in the field of science and technology